

ASSIGNMENT #7: INTEGRATED SPATIAL ANALYSIS AND ZONAL STATISTICS

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	7
TITLE	Integrated Spatial Analysis and Zonal Statistics
GRADE VALUE	9 points
DUE DATE	March 23, 2026, 11:59 PM PT

Objectives

After completing this assignment, you will be able to:

- Execute an integrated spatial analysis by layering multiple agricultural data types (vector and raster).
- Perform Zonal Statistics to extract numerical values from imagery (NDVI) into field boundaries.
- Quantify field performance by calculating average health metrics per polygon.
- Prepare a complex spatial visualization that demonstrates the intersection of soil, health, and boundaries.

Instructions

Step 1: Create Your Feature Branch

Open your terminal in VS Code and switch to a new branch for this integrated analysis:

- `git checkout -b feature/assignment-07-zonal-stats`

Step 2: Set Up Your Integration Notebook

01. Create a new notebook: notebooks/07_spatial_integration.ipynb.
02. Load your "Triple Threat" layers: * Your Field Boundaries (GeoJSON).
 - a. Your Soil Data (from Assignment #2/8).
 - b. Your NDVI Raster (from Assignment #5).
03. Coordinate Alignment: Double-check that all three layers are in the same CRS.

Mark complete: ☐

Step 3: Perform Zonal Statistics

Open your terminal in VS Code and switch to a new branch for this integrated analysis:

01. **Task:** Use a library like rasterstats or rioarray to calculate the mean NDVI for every field polygon in your dataset.
02. **Task:** Create a new column in your GeoDataFrame called mean_ndvi.
03. **Task:** Perform a spatial join to attach the soil health metrics to these same field boundaries.

Mark complete: ☐

Step 4: Identify Climate Anomalie

01. **Task:** Create a map that overlays at least 3 layers.
 - a. Example: Set the field transparency (alpha) so you can see the NDVI heatmap underneath, while using a specific border color to highlight fields with a certain soil type.

02. **Task:** Ensure the map includes a clear legend that explains all three variables.

Mark complete: ☐

Step 5: Design Your Dashboard Element

01. **Task:** Design 1 integrated spatial visualization that shows these overlapping data layers clearly for a non-technical user.
02. **Task:** Save this as
`output/dashboard_assets/integrated_spatial_analysis.png`.

Mark complete: ☐

Step 6: Update Your Project Tracker

Open `docs/project/assignment-01-tracker.md`.

- Document the most significant weather trend you observed.
- Note if any weather stations were missing data and how you handled those gaps (e.g., interpolation or using the nearest station).

Mark complete: ☐

Grading Criteria

Criterion	Points	Description
Multi-Layer Integration	3	Successfully combined Fields + Soil + NDVI into a single analytical environment.
Zonal Statistics	3	Correctly extracted average (mean) raster values into field polygons.
Integrated Design	2	Created 1 polished visualization showing overlapping data layers for the dashboard.
Technical Execution	1	Clean code and proper branch management.